# Sahaj Bhadja

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## Summary

Software Engineer with 2+ years building **C++/Python planning systems**, **autonomous-driving simulations**, and **reliable evaluation pipelines** using reinforcement learning and algorithms.

## Education

**University of Southern California, Los Angeles, CA** **August 2024 – May 2026**
*Master of Science in Computer Science* *GPA: 3.75/4*
Coursework: Database Systems, Analysis of Algorithms, Web Technologies

**Pandit Deendayal Energy University, Gandhinagar, India** **November 2020 – May 2024**
*Bachelor of Technology in Computer Science and Engineering* *GPA: 3.88/4*
Coursework: Operating Systems, Software Engineering, Object-Oriented Programming

## Experience

**USC Dornsife** | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant services** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **production data pipelines**, implementing validation, reconciliation, error handling, incremental processing, CI/CD releases, and stable monitoring workflows at enterprise scale.

**Grad8** | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent processing and sustained production traffic reliably.
- Maintained 90%+ scoring consistency across 10K+ submissions through **structured model evaluation**, model switching, fallback paths, persistent results, and reviewer controls supporting reproducible analytical workflows reliably for technical users.

**USC AD Labs** | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

**Aasma Technology Solutions** **August 2023 – July 2024**
*Software Developer Engineer - (SecureSync)* | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

**Pandit Deendayal Energy University** **January 2023 – May 2024**
*Research Fellowship* | *India*
- Reduced centralized-control and quantum-risk exposure by 25% through a **novel consensus algorithm**, comparing benchmark behavior, performance trade-offs, and failure characteristics against established blockchain mechanisms systematically and reproducibly.
- Integrated five lattice-based cryptographic primitives into a **quantum-resistant protocol**, documenting design decisions and benchmark evidence for documented peer-reviewed research presented at IISc Bengaluru through an IEEE publication.

## Skills

**Languages & Tools:** C++, Python, Bash, Git, GitHub, Linux
**Planning & ML:** Trajectory Planning, Decision Transformers, Offline RL, Simulation, PyTorch, Transformers, Attention
**Modeling:** Spatial-Temporal Models, HD Maps, Model Evaluation, Feature Engineering, Deep Learning, Data Validation
**Systems & Algorithms:** Data Structures, Algorithms, Complexity, OOP, Multithreading, Concurrency, Distributed Systems, CPU Scheduling
**Testing & Debugging:** Unit Testing, Integration Testing, Automated Testing, Regression Testing, PyTest, Debugging, Monitoring
**Engineering:** Performance Tuning, Fault Tolerance, Error Handling, Documentation, Agile, CI/CD

## Projects

**Off-Road Planning Evaluation Harness - Design Concept**
- Designed an **off-road planning evaluation harness** concept in C++ and Python with logged trajectory replay, terrain-aware scenarios, deterministic simulation, regression metrics, failure triage, and reproducible benchmark reports.

**CPU-Scheduling Simulator** | Live
- Built a **C++ CPU-scheduling simulator** implementing FCFS, SJF, SRTF, Priority, and Round Robin algorithms with configurable workloads, timeline visualization, comparative metrics, and deterministic validation across scheduling scenarios.

**Big-Brain** | Live
- Built an **AI-powered data platform** with secure authentication, real-time collaboration, semantic search, and ingestion across 50K+ documents, reducing review time by 60% through scalable automated workflows reliably.